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(54) SHEET CONVEYING DEVICE AND IMAGE FORMING APPARATUS

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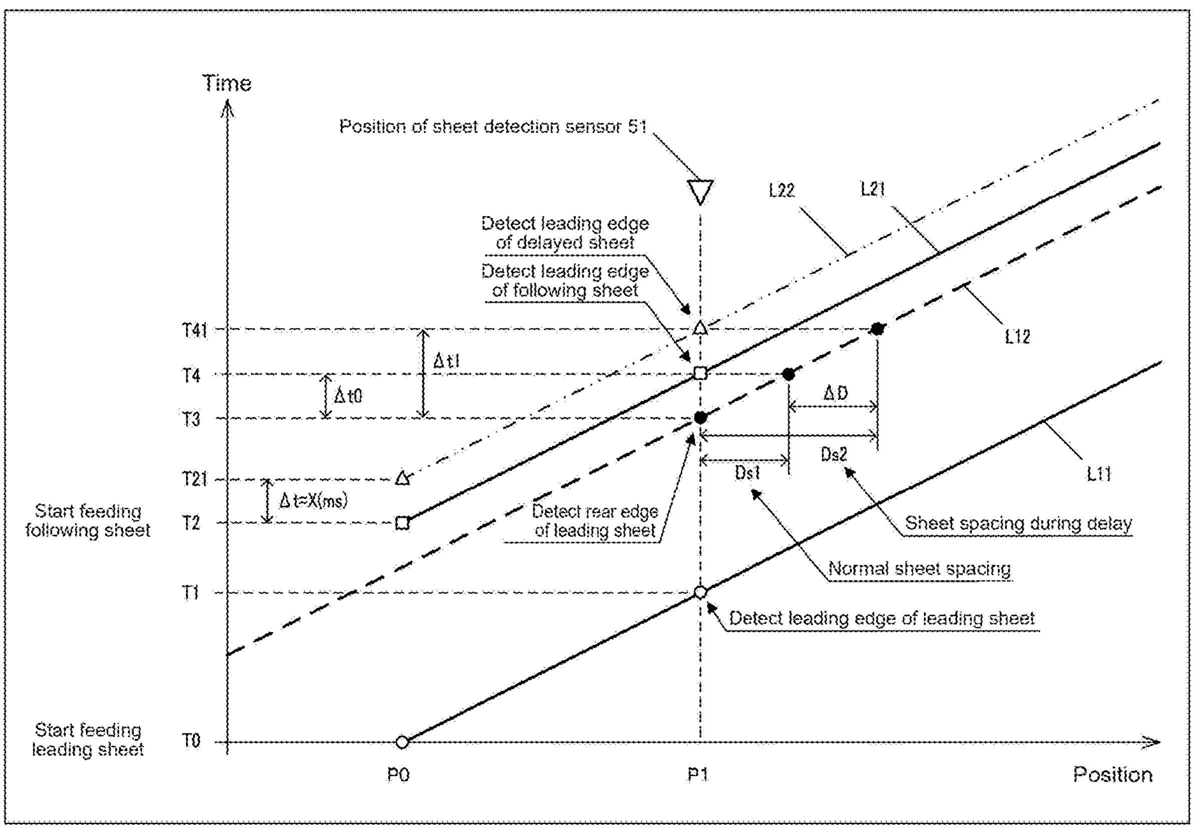
(57) ABSTRACT

A sheet conveying device includes: a first conveying roller; a second conveying roller; a sheet sensor; and a speed control unit that controls a conveying speed of a sheet by each of the first conveying roller and the second conveying roller, the speed control unit being configured to change, where a leading edge of a second sheet has not been detected at a reference time point, the conveying speed by the first conveying roller from an initial speed to a high speed faster than the initial speed, maintain the conveying speed by the second conveying roller at the initial speed, determine return timing on the basis of a second elapsed amount, and return the conveying speed by the first conveying roller from the high speed to the initial speed at the determined return timing.



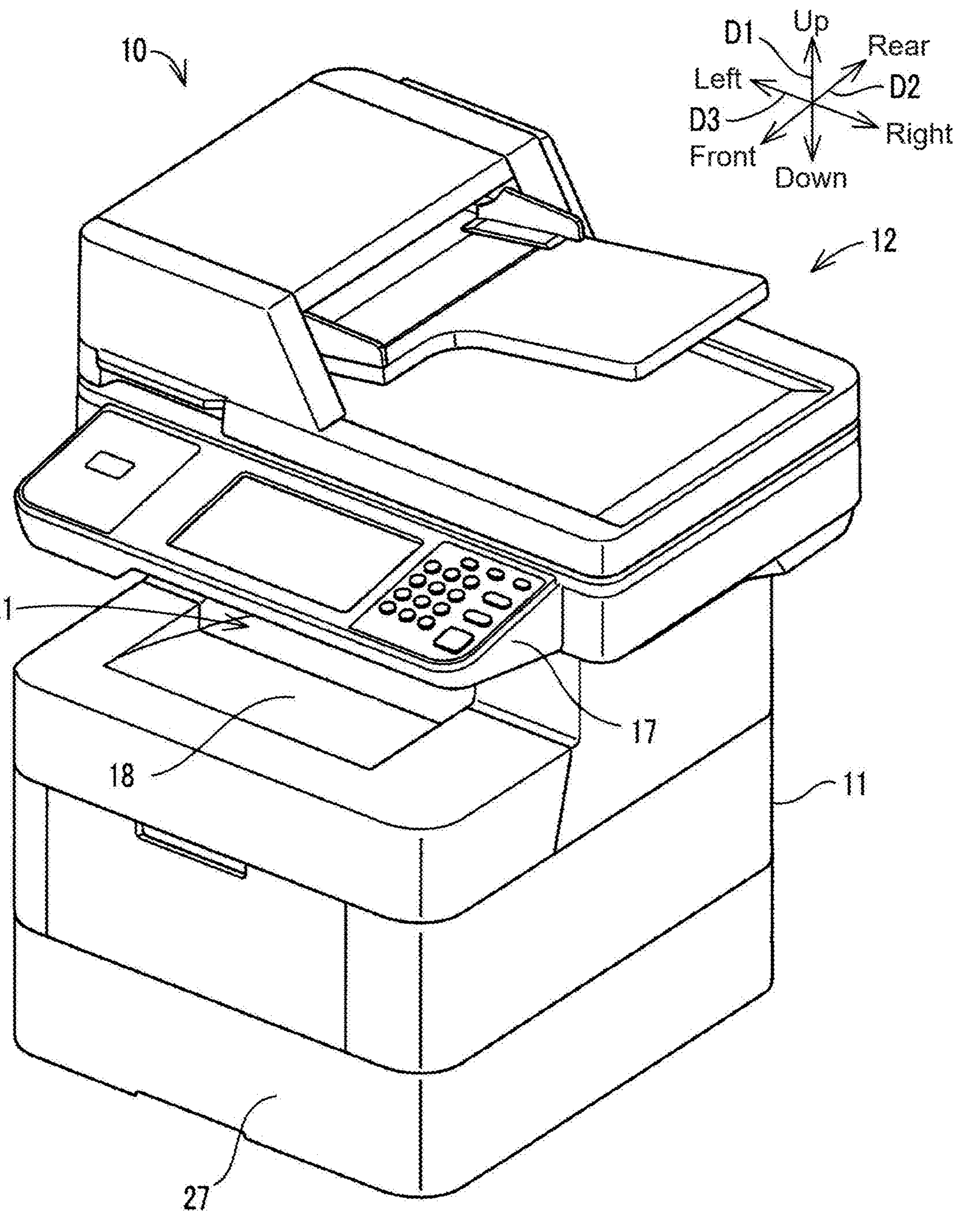


FIG.1

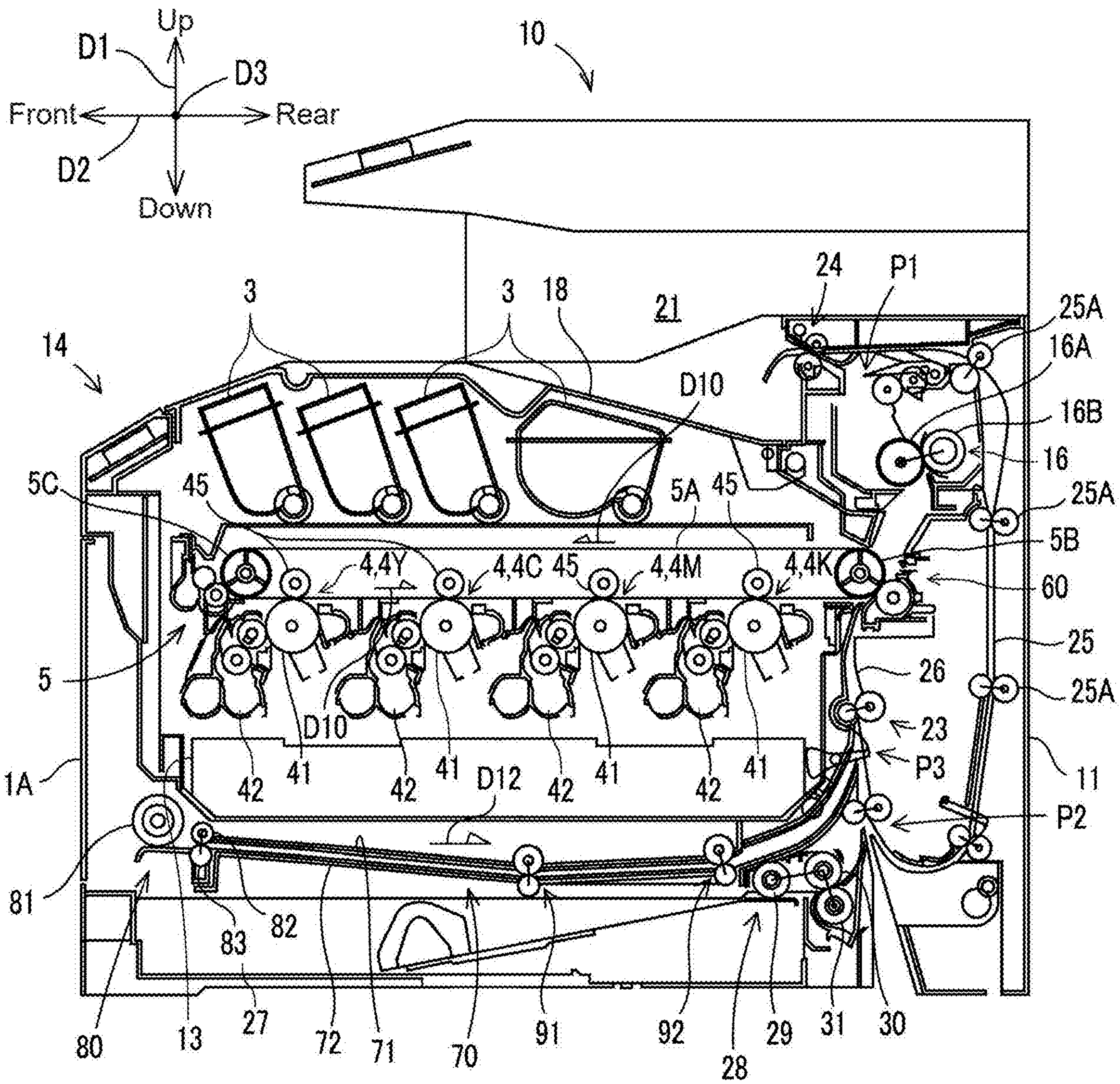

Up
D1
D3
Front
Rear
D2
Down
10
3
3
18
21
24
P1
25A
16A
16B
16
14
D10
5C
45
5A
45
25A
5B
4,4Y
4,4C
45
4,4M
4,4K
60
25
26
5
D10
23
25A
11A
42
41
42
41
42
41
42
41
P3
11
D12
P2
81
83
82
80
13
27
72
71
70
91
92
29
31
30
28

FIG.2

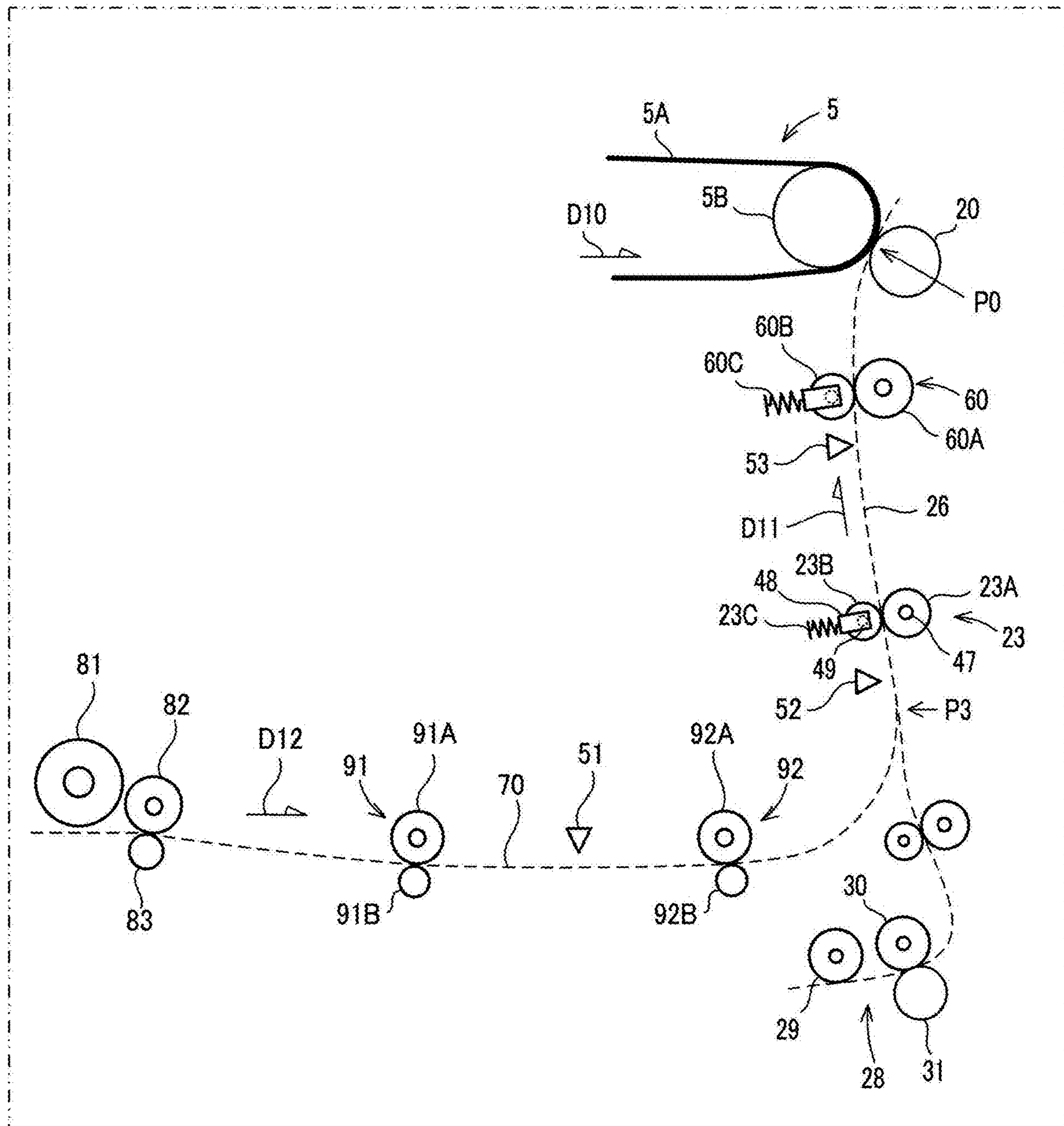

5A
5
5B
D10
20
P0
60B
60C
60
60A
53
D11
26
23B
48
23A
23C
23
49
47
52
P3
81
82
D12
91A
91
51
92A
92
70
83
91B
92B
30
29
28
31

FIG.3

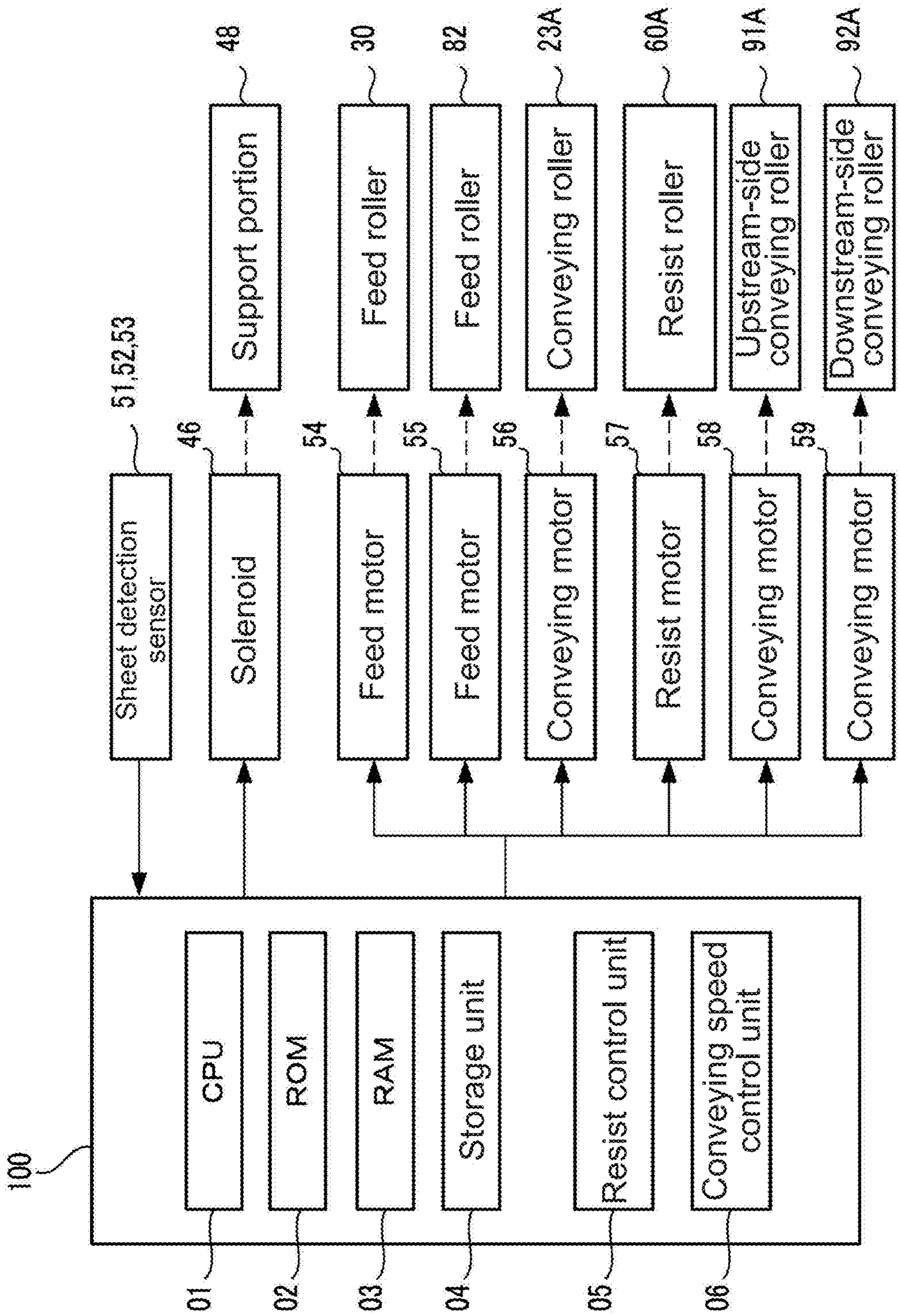
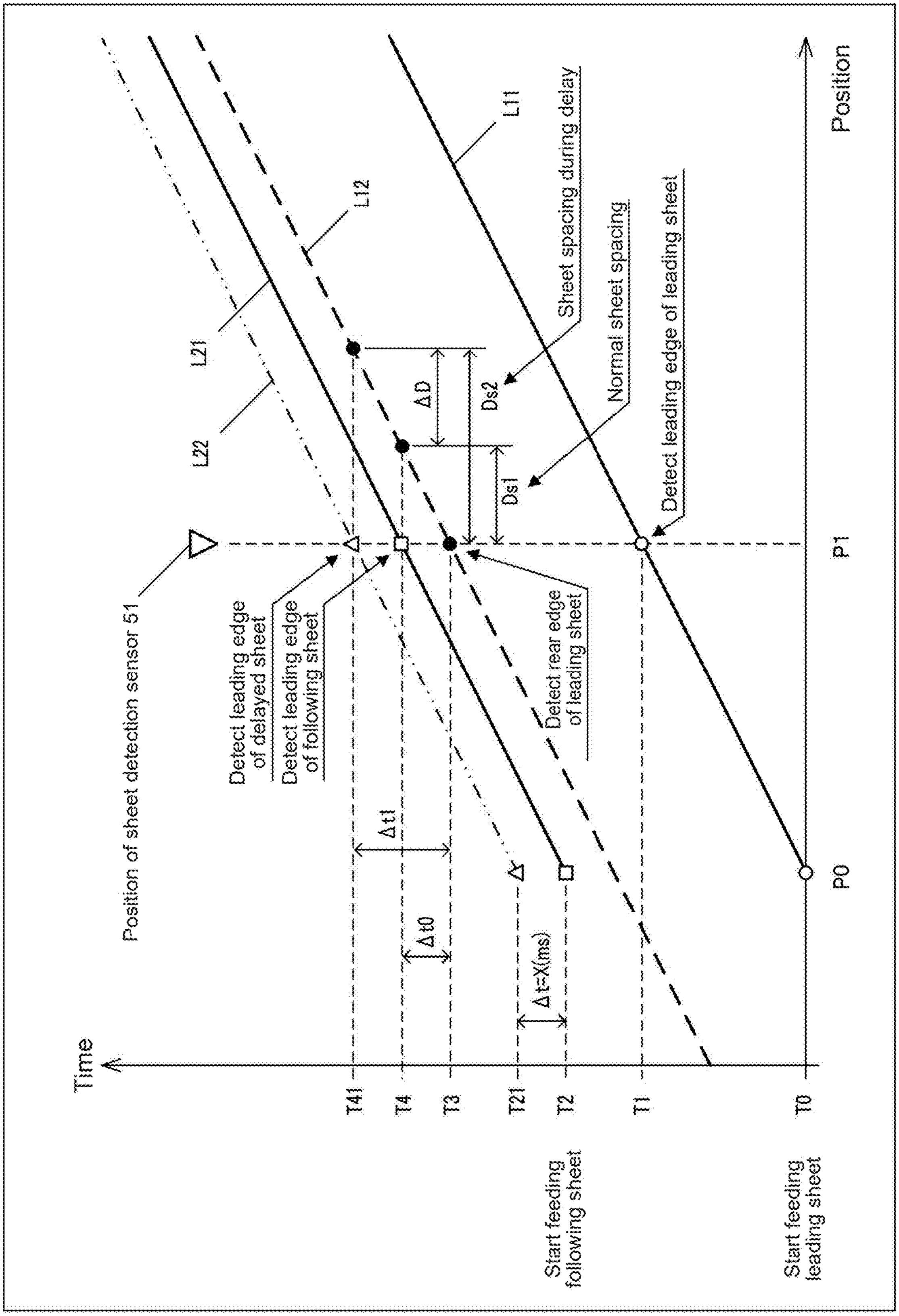

100
CPU
101
ROM
102
RAM
103
Storage unit
104
Resist control unit
105
Conveying speed control unit
106
Sheet detection sensor
51,52,53
Solenoid
46
Support portion
48
Feed motor
54
Feed roller
30
Feed motor
55
Feed roller
82
Conveying motor
56
Conveying roller
23A
Resist motor
57
Resist roller
60A
Conveying motor
58
Upstream-side conveying roller
91A
Conveying motor
59
Downstream-side conveying roller
92A

FIG.4



Time
Position
T0
T1
T2
T21
T3
T4
T41
P0
P1
Start feeding leading sheet
Start feeding following sheet
Position of sheet detection sensor 51
Detect leading edge of delayed sheet
Detect leading edge of following sheet
Detect rear edge of leading sheet
Detect leading edge of leading sheet
Normal sheet spacing
Sheet spacing during delay
L11
L12
L21
L22
Δt1
Δt0
Δt=X(ms)
ΔD
Ds1
Ds2

FIG.5

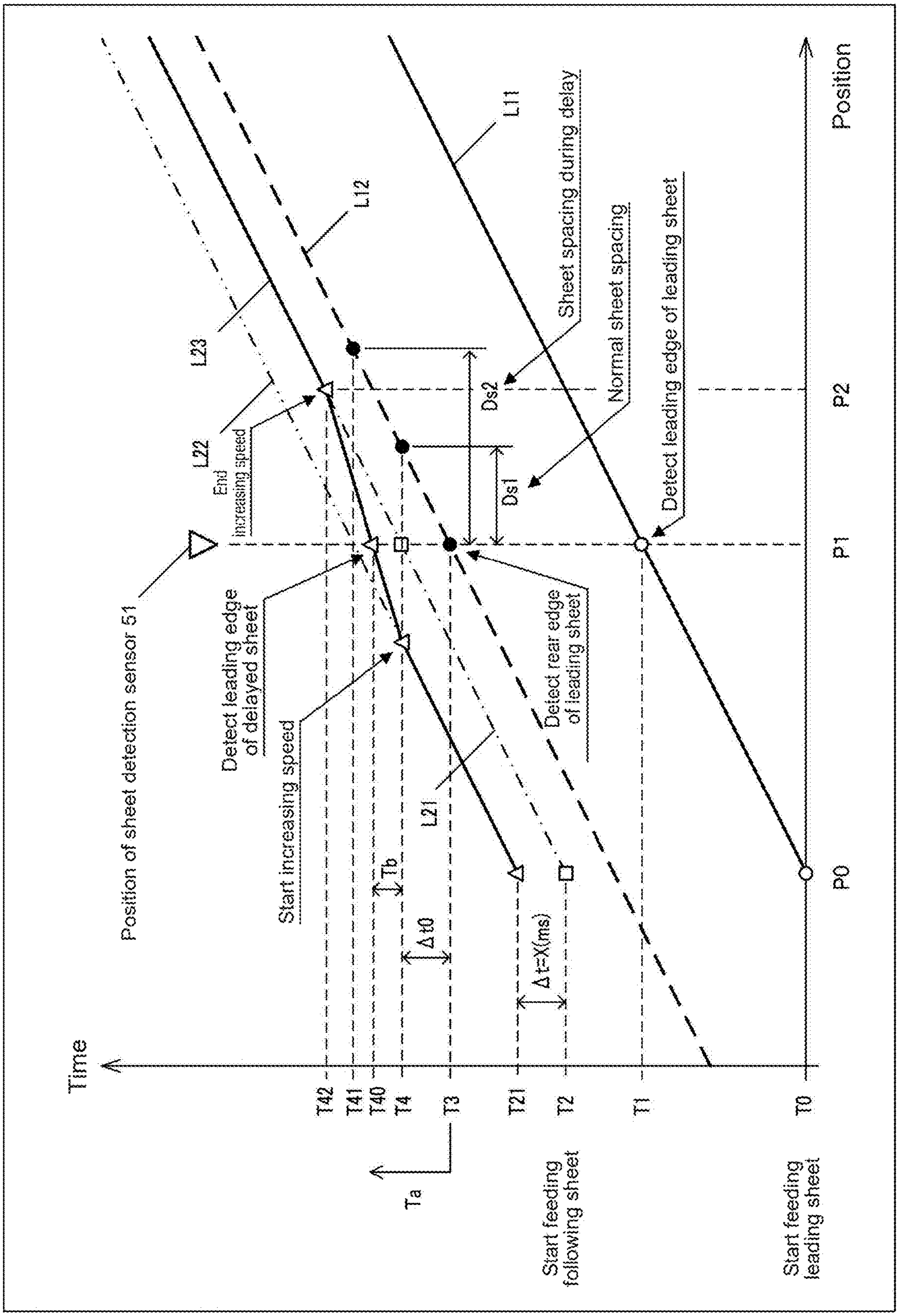

Time
Position
Position of sheet detection sensor 51
Start increasing speed
Detect leading edge
of delayed sheet
End
increasing speed
L22
L23
L12
L11
L21
Detect rear edge
of leading sheet
Sheet spacing during delay
Normal sheet spacing
Detect leading edge of leading sheet
Ds1
Ds2
Tb
Δt0
Δt=X(ms)
P0
P1
P2
T42
T41
T40
T4
T3
T21
T2
T1
T0
Ta
Start feeding
following sheet
Start feeding
leading sheet

FIG.6

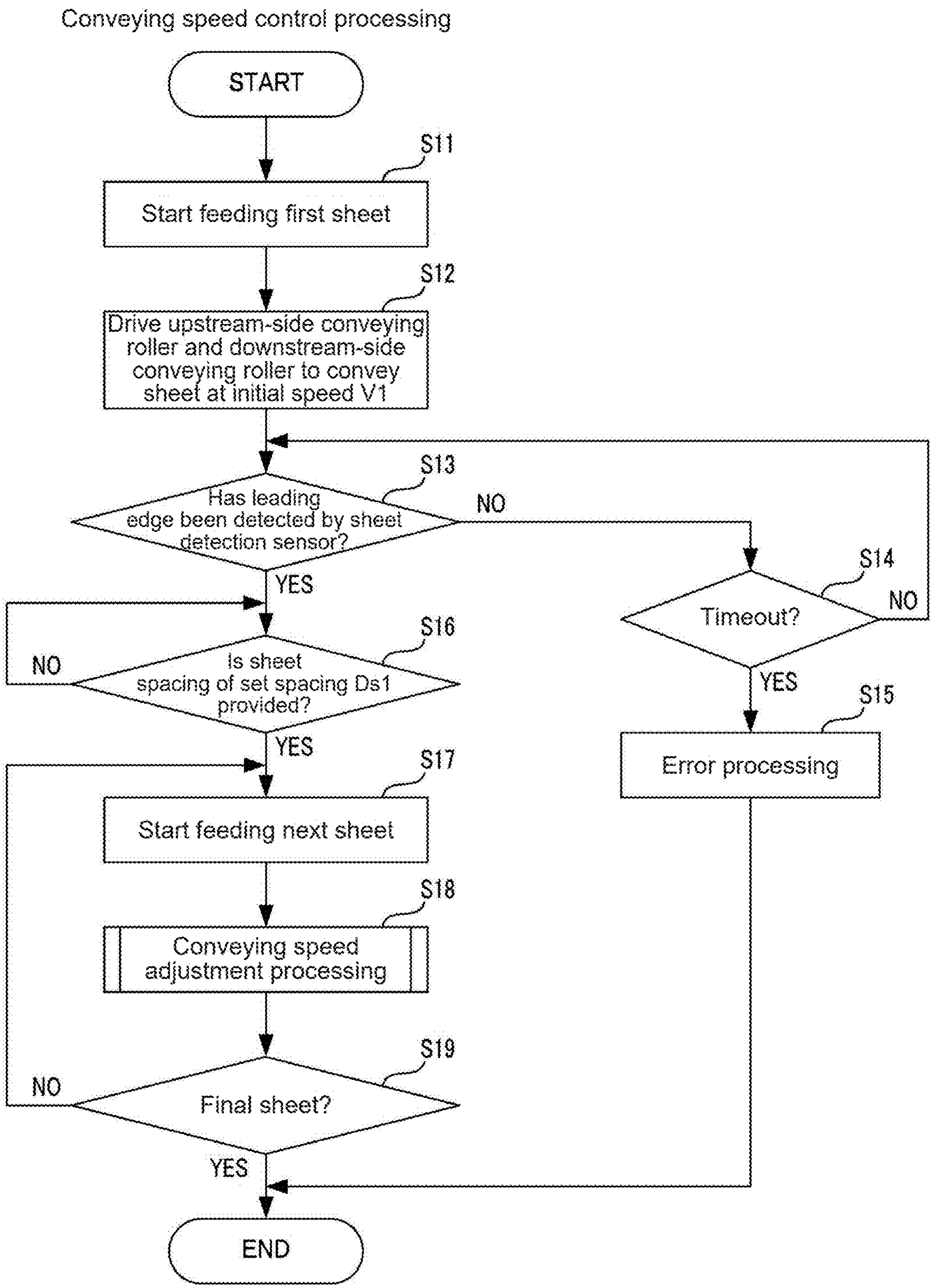

Conveying speed control processing
START
S11
Start feeding first sheet
S12
Drive upstream-side conveying roller and downstream-side conveying roller to convey sheet at initial speed V1
S13
Has leading edge been detected by sheet detection sensor?
NO
YES
S14
Timeout?
NO
YES
S15
Error processing
S16
Is sheet spacing of set spacing Ds1 provided?
NO
YES
S17
Start feeding next sheet
S18
Conveying speed adjustment processing
S19
Final sheet?
NO
YES
END

FIG.7

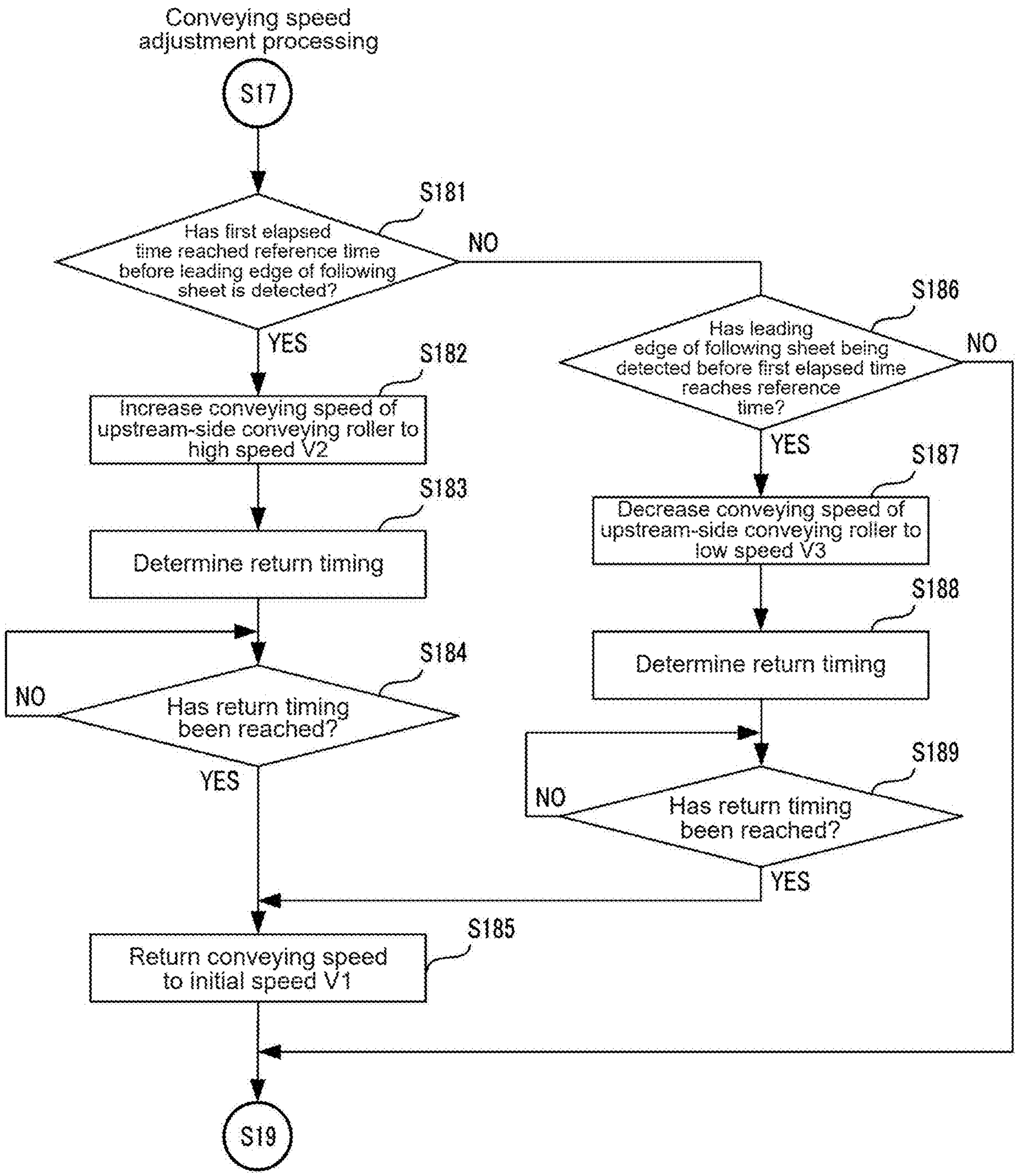

Conveying speed adjustment processing
S17
S181
Has first elapsed time reached reference time before leading edge of following sheet is detected?
NO
YES
S182
Increase conveying speed of upstream-side conveying roller to high speed V2
S183
Determine return timing
S184
Has return timing been reached?
NO
YES
S186
Has leading edge of following sheet being detected before first elapsed time reaches reference time?
NO
YES
S187
Decrease conveying speed of upstream-side conveying roller to low speed V3
S188
Determine return timing
S189
Has return timing been reached?
NO
YES
S185
Return conveying speed to initial speed V1
S19

FIG.8

SHEET CONVEYING DEVICE AND IMAGE FORMING APPARATUS

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2024-029750 filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a sheet conveying device capable of conveying a sheet and an image forming apparatus including this sheet conveying device.

BACKGROUND

[0003] Image forming apparatuses such as a printer, a copier, a facsimile machine, and a multifunction device having these functions are provided with a resist roller unit including a resist roller pair. The resist roller unit stands by while the leading edge of a sheet is pressed against the nip portion of the resist roller pair being stopped to correct the inclination of the sheet and then feeds out the sheet to an image transfer position by causing the resist roller pair to be driven to rotate at predetermined feed timing that matches transfer start timing at which the transfer of an image to the sheet starts at the image transfer position.

[0004] Further, the image forming apparatus is provided with a sheet conveying device that conveys a sheet from a sheet housing unit toward the resist roller unit.

[0005] Further, for example, there is known a sheet supplying device that controls the spacing between the rear edge of the leading sheet and the leading edge of the next following sheet by changing the timing at which the conveying speed of the sheet decelerates to the conveying speed of the sheet at the image transfer position.

SUMMARY

[0006] A sheet conveying device according to an aspect of the present disclosure is a sheet conveying device that sequentially conveys a plurality of sheets, including: a first conveying roller that is driven to rotate by receiving driving force from a first drive unit to convey a sheet; a second conveying roller that is provided downstream of the first conveying roller in a sheet conveying direction and is driven to rotate by receiving driving force from a second drive unit to convey the sheet; a sheet sensor that is provided between the first conveying roller and the second conveying roller and detects a leading edge and a rear edge of the sheet; and a speed control unit that controls a conveying speed of the sheet by each of the first conveying roller and the second conveying roller. The speed control unit is configured to change, where a leading edge of a following second sheet has not been detected at a reference time point at which a first elapsed amount from a first time point at which a rear edge of a first sheet that is fed previously is detected reaches a reference amount corresponding to predetermined sheet spacing, the conveying speed of the sheet by the first conveying roller from an initial speed to a high speed faster than the initial speed, and maintain the conveying speed of the sheet by the second conveying roller at the initial speed, determine, on a basis of a second elapsed amount from the reference time point to a second time point at which the leading edge of the second sheet is detected, return timing at which the conveying speed of the sheet by the first conveying roller is returned from the high speed to the initial speed, and return the conveying speed of the sheet by the first conveying roller from the high speed to the initial speed at the determined return timing.

[0007] An image forming apparatus according to another aspect of the present disclosure includes: the sheet conveying device, a toner image being transferred to a sheet conveyed by the sheet conveying device to an image transfer position.

[0008] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF FIGURES

[0009] FIG. **1** is a perspective view showing a configuration of an image forming apparatus according to an embodiment of the present disclosure.

[0010] FIG. **2** is a schematic diagram showing an internal configuration of the image forming apparatus.

[0011] FIG. **3** is a schematic diagram showing a configuration of the periphery of a conveying path of the image forming apparatus.

[0012] FIG. **4** is a block diagram showing a configuration of the image forming apparatus.

[0013] FIG. **5** is a graph showing a relationship between the position of each sheet and the arrival time point in the case where sheets are continuously conveyed from a sheet receiving unit at a constant conveying speed.

[0014] FIG. **6** is a graph showing a relationship between the position of each sheet and the arrival time point in the case where the conveying speed of the following sheet conveyed from the sheet receiving unit has been changed.

[0015] FIG. **7** is a flowchart showing an example of the procedure of conveying speed control processing executed by a control unit of the image forming apparatus.

[0016] FIG. **8** is a flowchart showing an example of the procedure of conveying speed adjustment processing executed by the control unit of the image forming apparatus.

DETAILED DESCRIPTION

[0017] Embodiments of the present disclosure will be described below with reference to the drawings as appropriate. The embodiments described below are merely examples embodying the present disclosure and do not limit the technical scope of the present disclosure.

[0018] FIG. **1** is a perspective view showing a configuration of an image forming apparatus **10** according to the embodiment of the present disclosure. FIG. **2** is a schematic diagram showing an internal configuration of the image forming apparatus **10**. In FIG. **2**, illustration of an image reading unit **12** is omitted. In the following description, an up-and-down direction D**1** is defined with reference to the state in which the image forming apparatus **10** is installed to be ready for use (state shown in FIG. **1**), a front-and-rear direction D2 is defined with the front side (front surface side) of the image forming apparatus **10** as the front, and a right-and-left direction D3 is defined with the image forming apparatus **10** viewed from the front side (front surface side).

[Image Forming Apparatus **10**]

[0019] As shown in FIG. **1**, the image forming apparatus **10** is a multifunction device capable of printing an image on a sheet such as printing paper and has various functions such as a printing function, a copying function, a facsimile function, and a scanning function. The image forming apparatus **10** is not limited to the multifunction device and only needs to be an apparatus having a printing function of printing an image on the conveyed sheet. For example, the image forming apparatus **10** may be a printer, a copier, or a FAX machine.

[0020] The image forming apparatus **10** includes the image reading unit **12** and an image forming unit **14**. The image reading unit **12** performs image reading processing of reading an image of a document and is provided in the upper part of the image forming apparatus **10**. The image forming unit **14** performs image forming processing of forming a color image on the basis of an electrophotographic method and is provided in the lower part of the image forming apparatus **10**.

[0021] An output space **21** is provided above the image forming unit **14**. The sheet on which the image has been formed is output from a sheet output port **15**A (see FIG. **2**) to the output space **21**.

[0022] The image forming unit **14** includes a casing **11** as a body of an apparatus. The respective units constituting the image forming unit **14** are disposed inside the casing **11**. The casing **11** includes an outer frame that covers the entire image forming unit **14** and an inner frame for supporting the respective units constituting the image forming unit **14**.

[0023] The image forming unit **14** forms a color image on a sheet such as printing paper on the basis of a so-called tandem method.

[0024] As shown in FIG. **2**, the image forming unit **14** includes a plurality of image formation units **4**, an intermediate transfer unit **5**, a light scanning device **13**, a secondary transfer roller **20**, a fixing device **16**, a sheet tray **18**, a sheet housing unit **27**, a first feeding unit **28**, an operation display unit **17** (see FIG. **1**), a sheet conveying path **26** (hereinafter, abbreviated as a conveying path **26**), a sheet conveying unit **23**, a resist roller unit **60**, a toner container **3**, a multi-sheet conveying path **70** (hereinafter, abbreviated as a conveying path **70**), a second feeding unit **80**, a control unit **100** (see FIG. **4**), and the like. Note that the image forming unit **14** may form a monochrome image on a sheet by one image formation unit **4**. In this case, the intermediate transfer unit **5** does not necessarily need to be provided.

[0025] As shown in FIG. **2**, the sheet housing unit **27** is provided at the lowest part of the image forming apparatus **10**. The sheet housing unit **27** houses a plurality of sheets and is formed in, for example, a tray shape with the top opened. The sheet housing unit **27** is supported by the casing **11**.

[0026] The first feeding unit **28** takes a plurality of sheets stacked in the sheet housing unit **27** one sheet at a time and sequentially feeds the taken sheet toward the conveying path **26**. The first feeding unit **28** includes a pick-up roller **29**, a feed roller **30**, and a separation roller **31**. The pick-up roller **29** and the feed roller **30** are provided above the rear portion of the sheet housing unit **27**. The separation roller **31** is provided on the lower side of the feed roller **30** while being in contact with the roller surface of the feed roller **30**. Since the separation roller **31** is provided, even if a plurality of sheets is picked up by the pick-up roller **29**, only the topmost sheet of the fed sheets is separated from the other sheets by the separation roller **31**.

[0027] The feed roller **30** is connected to a feed motor **54** (see FIG. **4**). The feed roller **30** conveys the sheet to the downstream side in the conveying direction D**11** by receiving the rotational driving force from the feed motor **54**. A drive transmission mechanism (not shown) that transmits the rotation of the feed roller **30** to the pick-up roller **29** is provided between the feed roller **30** and the pick-up roller **29**. The pick-up roller **29** and the feed roller **30** are connected via the drive transmission mechanism. When the feed roller **30** is caused to rotate by the feed motor **54**, the pick-up roller **29** also rotates in the same direction and at the same circumferential speed as those of the feed roller **30** by the drive transmission mechanism.

[0028] When an instruction signal for feeding a sheet in the sheet housing unit **27** is input to the image forming apparatus **10**, the feed roller **30** and the pick-up roller **29** are caused to rotate by the rotational driving force of the feed motor **54**, and the sheet is fed from the sheet housing unit **27** to the conveying path **26**. Specifically, a sheet in the sheet housing unit **27** is taken by the pick-up roller **29** and fed out to the downstream side in the feed direction, and when the tip portion of the sheet reaches the nip portion between the feed roller **30** and the separation roller **31**, the feed roller **30** conveys the sheet to the conveying path **26**.

[0029] The conveying path **26** is a guide path that guides the sheet fed by the feed roller **30** to the sheet output port **15**A. As shown in FIG. **2**, the conveying path **26** is curved upward from the feed roller **30** and then extends upward. The conveying path **26** passes through the secondary transfer roller **20** and reaches the sheet output port **15**A.

[0030] The sheet conveying unit **23** and the resist roller unit **60** are provided in the conveying path **26**.

[0031] The sheet conveying unit **23** conveys the sheet fed to the conveying path **26** by the first feeding unit **28** in the conveying direction D**11** toward an image transfer position P**0**. The image transfer position P**0** is a position where a toner image on a transfer belt **5**A is transferred to the sheet and is a position where a drive roller **5**B and the secondary transfer roller **20** face each other. The sheet conveying unit **23** conveys the sheet to the downstream side in the conveying direction D**11** by receiving the rotational driving force from the conveying motor **56** (see FIG. **4**). The configuration of the sheet conveying unit **23** will be described below.

[0032] The resist roller unit **60** is disposed upstream of the image transfer position P**0** in the conveying direction D**11** and downstream of the sheet conveying unit **23** in the conveying direction D**11**, in the conveying path **26**. That is, the resist roller unit **60** is provided between the image transfer position P**0** and the sheet conveying unit **23**. The resist roller unit **60** corrects the inclination of the sheet conveyed through the conveying path **26** while being inclined with respect to the conveying direction D**11** and conveys the corrected sheet to the downstream side in the conveying direction D**11**. The configuration of the resist roller unit **60** will be described below.

[0033] As shown in FIG. **2**, each of the image formation units **4** is provided below the intermediate transfer unit **5**.

Each image formation unit **4** performs image forming processing of forming a toner image on the surface of the transfer belt **5**A on the basis of image data input from the outside. The plurality of image formation units **4** is arranged along the travelling direction of the transfer belt **5**A (direction indicated by an arrow D**10**). From the front side to the rear side of the transfer belt **5**A, an image formation unit **4**Y for yellow, an image formation unit **4**C for cyan, an image formation unit **4**M for magenta, and an image formation unit **4**K for black are arranged in the stated order in a single line.

[0034] Each of the image formation units **4** includes a photoreceptor drum **41**, a charging device **42**, a development device **44**, a primary transfer roller **45**, and the like. The image formation unit **4**Y forms a toner image on the surface of the photoreceptor drum **41** using a yellow toner. The image formation unit **4**C, the image formation unit **4**M, and the image formation unit **4**K respectively form toner images on the surface of the photoreceptor drum **41** with a cyan toner, a magenta toner, and a black toner. The processing of developing the toner image on the photoreceptor drum **41** is performed by the development device **44**.

[0035] The intermediate transfer unit **5** includes the transfer belt **5**A, the drive roller **5**B, and a driven roller **5**C. The transfer belt **5**A is a belt member to which toner images of respective colors, which are formed on the photoreceptor drums **41** of the image formation units **4**, are transferred. The transfer belt **5**A is provided above the photoreceptor drum **41**. The transfer belt **5**A is an endless circular belt. The transfer belt **5**A is rotatably supported by the drive roller **5**B and the driven roller **5**C provided spaced apart in the right-and-left direction D**3**. The transfer belt **5**A is supported by being stretched over the drive roller **5**B and the driven roller **5**C. When the surface of the transfer belt **5**A passes between the photoreceptor drum **41** and the primary transfer roller **45**, the toner image is transferred in order from each photoreceptor drum **41** in a superimposed manner. The toner image transferred to the transfer belt **5**A is conveyed to the image transfer position P**0** and is transferred to the sheet at the image transfer position P**0**.

[0036] The light scanning device **13** applies laser light to the photoreceptor drum **41** of each image formation unit **4** on the basis of the input image data of each color. This forms an electrostatic latent image on each photoreceptor drum **41**. When an image forming instruction to form images on a plurality of sheets and pieces of image data corresponding to the respective sheets are input to the image forming apparatus **10**, the control unit **100** determines scanning start timing at which laser scanning on the photoreceptor drum **41** with laser light based on the image data is started and applies the laser light to the photoreceptor drum **41** at the scanning start timing determined for each of the plurality of sheets.

[0037] The secondary transfer roller **20** is provided to face the drive roller **5**B with the conveying path **26** extending vertically sandwiched therebetween. The secondary transfer roller **20** performs transfer processing of transferring the toner image on the transfer belt **5**A to the sheet by the transfer potential applied to the secondary transfer roller **20**. The position where the secondary transfer roller **20** transfers the toner image to the sheet is the image transfer position P**0**. The sheet to which the toner image has been transferred is conveyed to the fixing device **16**.

[0038] The fixing device **16** heats the toner image transferred to the sheet to fix the toner image to the sheet and includes a heating roller **16**A and a pressure roller **16**B. The sheet conveyed to the fixing device **16** is conveyed while being sandwiched between the heating roller **16**A and the pressure roller **16**B. During this conveyance, heat is transmitted from the heating roller **16**A to the toner image transferred to the sheet to heat the toner image. This fixes the toner image to the sheet.

[0039] The sheet that has passed through the fixing device **16**, on which the image has been formed, is output from the sheet output port **15**A to the sheet tray **18** by an output roller **24** provided at the most downstream end of the conveying path **26**.

[0040] As shown in FIG. **2**, the conveying path **70** is provided between the light scanning device **13** and the sheet housing unit **27**. The conveying path **70** is a guide path that guides the sheet fed by the second feeding unit **80** described below into the conveying path **26**. The conveying path **70** is formed by an upper guide member **71** and a lower guide member **72**, which extend in the front-and-rear direction D**2**. The upper guide member **71** and the lower guide member **72** are disposed to be spaced apart at a predetermined distance so as to face each other in the up-and-down direction D**1**. The space sandwiched between the upper guide member **71** and the lower guide member **72** is the conveying path **70**. The conveying path **70** extends in the front-and-rear direction D**2** of the image forming unit **14** and is connected to the conveying path **26** at a connection point P**3** (see FIG. **3**) on the rear side the image forming unit **14**.

[0041] A manual sheet receiving unit **11**A is provided on the front surface side of the image forming unit **14**. The sheet receiving unit **11**A serves also as a cover for the front surface of the casing **11** of the image forming unit **14**.

[0042] The sheet receiving unit **11**A is configured to be capable of opening/closing the entrance of the conveying path **70** on the front surface of the casing **11**. FIG. **2** shows the state in which the sheet receiving unit **11**A is closed against the front surface of the casing **11**. When the sheet receiving unit **11**A is opened against the front surface of the casing **11** and the inner surface thereof is turned upward, a plurality of sheets having an arbitrary size can be placed on the inner surface.

[0043] The second feeding unit **80** is provided near the entrance on the front side of the conveying path **70**. The second feeding unit **80** takes a plurality of sheets placed on the sheet receiving unit **11**A one sheet at a time and sequentially feeds the taken sheet toward the conveying path **70**. The second feeding unit **80** includes a pick-up roller **81**, a feed roller **82**, and a driven roller **83**.

[0044] The feed roller **82** is connected to a feed motor **55** (see FIG. **4**). The feed roller **82** conveys the sheet to the downstream side in a conveying direction D**12** by receiving the rotational driving force of the feed motor **55**. A drive transmission mechanism (not shown) that transmits the rotation of the feed roller **82** to the pick-up roller **81** is provided between the feed roller **82** and the pick-up roller **81**. The pick-up roller **81** and the feed roller **82** are connected via the drive transmission mechanism. When the feed roller **82** is caused to rotate by the feed motor **55**, the pick-up roller **81** also rotates in the same direction as that of the feed roller **82** by the drive transmission mechanism.

[0045] When an instruction signal for feeding a sheet in the sheet receiving unit **11**A is input to the image forming apparatus **10**, the feed roller **82** and the pick-up roller **81** are caused to rotate by the rotational driving force of the feed motor **55**, and the sheet is fed from the sheet receiving unit

11A to the conveying path 70. Specifically, a sheet in the sheet receiving unit 11A is taken by the pick-up roller 81 and fed out to the downstream side in the conveying direction D12, and when the tip portion of the sheet reaches the nip portion between the feed roller 82 and the driven roller 83, the feed roller 82 conveys the sheet to the conveying path 70.

[0046] A first conveying roller pair 91 and a second conveying roller pair 92 are provided in the conveying path 70. The first conveying roller pair 91 and the second conveying roller pair 92 convey the sheet fed to the conveying path 70 by the second feeding unit 80 in the conveying direction D12 to the conveying path 26. The configurations of the first conveying roller pair 91 and the second conveying roller pair 92 will be described below.

[0047] In the image forming unit 14, in the case where an image is formed on both surfaces of the sheet, the sheet that has passed through the fixing device 16 and has one surface on which the image has been formed is turned over and then conveyed to the upstream side of the secondary transfer roller 20 again. In detail, the leading edge of the sheet that has one surface on which the image has been formed is exposed from the sheet output port 15A to the outside, and the output rollers 24 are stopped in this state. At this time, the rear edge of the sheet is held while being sandwiched between the output rollers 24. After that, when the output rollers 24 are driven in the reverse direction, the sheet is switched back and fed backward.

[0048] As shown in FIG. 2, in the image forming unit 14, a reverse conveying path 25 that branches off from a branch point P1 defined on the downstream side of the fixing device 16 in the conveying path 26 and is connected to a merging point P2 defined in the conveying path 26 is formed. The merging point P2 is defined upstream of the sheet conveying unit 23 in the conveying path 26. The sheet that has been fed backward from the sheet output port 15A is guided into the reverse conveying path 25, is conveyed by a conveying roller 25A provided in the reverse conveying path 25, merges into the conveying path 26 through the reverse conveying path 25, and is conveyed to the secondary transfer roller 20 again. After that, a toner image is transferred to the back surface of the sheet that has reached the secondary transfer roller 20, and the image is formed on the back surface of the sheet by passing through the fixing device 16. The sheet with the images formed on both surfaces thereof is output from the sheet output port 15A to the sheet tray 18 by the output roller 24 that has been returned to be driven in the forward direction.

[Sheet Conveying Unit 23]

[0049] As shown in FIG. 3, the sheet conveying unit 23 includes a conveying roller 23A that is driven to rotate by receiving the driving force from the conveying motor 56 (see FIG. 4) and a driven roller 23B that is disposed while being in contact with the outer peripheral surface of the conveying roller 23A. The driving force from the conveying motor 56 is transmitted to the rotation shaft 47 of the conveying roller 23A. The conveying roller 23A and the driven roller 23B realize a conveying roller pair.

[0050] A rotation shaft 49 of the driven roller 23B is biased toward the conveying roller 23A by a spring 23C with predetermined elastic force (spring force). This causes the driven roller 23B to be pressed against the conveying roller 23A. When the conveying roller 23A is driven to rotate in this state, the driven roller 23B is driven in accordance therewith.

[0051] The rotation shaft 49 of the driven roller 23B is supported by a support portion 48 provided in the guide member of the conveying path 26, or the like. In this embodiment, the driven roller 23B is supported by the support portion 48 such that it is movable between the contact position where it is in contact with the conveying roller 23A and the contact release position where it is separated from the conveying roller 23A.

[0052] FIG. 4 is a block diagram showing a configuration of the image forming apparatus 10. A solenoid 46 is provided inside the casing 11. The solenoid 46 is connected to the control unit 100 and operates when energized by the control unit 100. The plunger of the solenoid 46 is connected to the support portion 48 via a linkage member (not shown). When the solenoid 46 is energized, the plunger operates to cause the support portion 48 to move from the contact position to the contact release position. When the solenoid 46 is then de-energized, the plunger is returned to the original position by an extension spring provided in the solenoid 46 and the support portion 48 is returned to the contact position by the spring force of the spring 23C.

[0053] As shown in FIG. 3, sheet detection sensors 52 and 53 are provided in the conveying path 26. The sheet detection sensors 52 and 53 detect the leading edge or rear edge of the sheet being conveyed. The sheet detection sensors 52 and 53 are provided near the center of the conveying path 26 in the width direction.

[0054] The sheet detection sensor 52 is provided upstream of the sheet conveying unit 23 in the conveying direction D11 and downstream of the connection point P3 between the conveying path 70 and the conveying path 26, in the conveying path 26. That is, the sheet detection sensor 52 is provided between the sheet conveying unit 23 and the connection point P3. In this embodiment, the sheet detection sensor 52 is provided at a position proximate to the sheet conveying unit 23.

[0055] The sheet detection sensor 53 is provided upstream of the resist roller unit 60 in the conveying direction D11 and downstream of the sheet conveying unit 23, in the conveying path 26. That is, the sheet detection sensor 53 is provided between the resist roller unit 60 and the sheet conveying unit 23. In this embodiment, the sheet detection sensor 53 is provided at a position proximate to the resist roller unit 60.

[Resist Roller Unit 60]

[0056] As shown in FIG. 3, the resist roller unit 60 is provided in the conveying path 26. The resist roller unit 60 is provided upstream of the image transfer position P0 in the conveying direction D11 and downstream of the sheet detection sensor 53 in the conveying direction D11, in the conveying path 26.

[0057] The resist roller unit 60 corrects the inclination (convey deviation) of the sheet conveyed while being inclined with respect to the conveying direction D11 in the sheet conveying unit 23 and conveys the corrected sheet to the image transfer position P0 at predetermined timing.

[0058] As shown in FIG. 3, the resist roller unit 60 includes a resist roller 60A that is driven to rotate by receiving the driving force from a resist motor 57 (see FIG. 4) and a driven roller 60B that is disposed while being in contact with the outer peripheral surface of the resist roller

60A. The driving force from the resist motor 57 is transmitted to the rotation shaft of the resist roller 60A. The resist roller 60A and the driven roller 60B realize a conveying roller pair.

[0059] The driven roller 60B is biased toward the resist roller 60A by a spring 60C. This causes the driven roller 60B to be pressed against the resist roller 60A. When the resist roller 60A is driven to rotate in this state, the driven roller 60B is driven in accordance therewith.

[First Conveying Roller Pair 91]

[0060] In the conveying path 70, the first conveying roller pair 91 is provided downstream of the second feeding unit 80 in the conveying direction D12. The first conveying roller pair 91 includes an upstream-side conveying roller 91A (an example of the first conveying roller according to the present disclosure) that is driven to rotate by receiving the driving force from a conveying motor 58 (an example of the first drive unit according to the present disclosure, see FIG. 4) and a driven roller 91B that is disposed while being in contact with the outer peripheral surface of the upstream-side conveying roller 91A. The driving force from the conveying motor 58 is transmitted to the rotation shaft of the upstream-side conveying roller 91A. The driven roller 91B is biased toward the upstream-side conveying roller 91A by, for example, a spring (not shown).

[Second Conveying Roller Pair 92]

[0061] In the conveying path 70, the second conveying roller pair 92 is provided downstream of the first conveying roller pair 91 in the conveying direction D12 and upstream of the connection point P3 in the conveying direction D12. The second conveying roller pair 92 includes a downstream-side conveying roller 92A (an example of a second conveying roller according to the present disclosure) that is driven to rotate by receiving the driving force from a conveying motor 59 (an example of a second drive unit according to the present disclosure, see FIG. 4) and a driven roller 92B that is disposed while being in contact with the outer peripheral surface of the downstream-side conveying roller 92A. The driving force from the conveying motor 59 is transmitted to the rotation shaft of the downstream-side conveying roller 92A. The driven roller 92B is biased toward the downstream-side conveying roller 92A by, for example, a spring (not shown).

[0062] As shown in FIG. 3, a sheet detection sensor 51 (an example of a sheet sensor according to the present disclosure) is provided in the conveying path 70. The sheet detection sensor 51 detects the leading edge or rear edge of the sheet being conveyed. The sheet detection sensor 51 is provided near the center of the conveying path 70 in the width direction. The sheet detection sensor 51 is provided downstream of the first conveying roller pair 91 in the conveying direction D12 and upstream of the second conveying roller pair 92, in the conveying path 70. That is, the sheet detection sensor 51 is provided between the first conveying roller pair 91 and the second conveying roller pair 92. In this embodiment, for example, the sheet detection sensor 51 is provided at substantially the midpoint between the first conveying roller pair 91 and the second conveying roller pair 92.

[0063] Each of the sheet detection sensors 51 to 53 is, for example, a reflective optical sensor. Each of the sheet detection sensors 51 to 53 is connected to the control unit 100 and the detection signal thereof is transmitted to the control unit 100. The control unit 100 detects the leading edge or rear edge of the conveyed sheet on the basis of the change in the detection signal transmitted from each of the sheet detection sensors 51 to 53. Note that since such a detection method has been known from the past, detailed description thereof is omitted.

[0064] Each of the feed motors 55 to 59 is, for example, a stepping motor or an inner brushless motor. These motors have high speed response and high position accuracy, but requires a certain waiting time before re-driving after stopping.

[Control Unit 100]

[0065] The control unit 100 controls the image forming apparatus 10, controls the rotational driving of each roller provided in the conveying paths 26 and 70, and controls the conveying speed of the sheet by each roller.

[0066] As shown in FIG. 4, the control unit 100 includes a CPU 101, a ROM 102, a RAM 103, a storage unit 104, and the like. The control unit 100 is electrically connected to the respective motors 54 to 59, the sheet detection sensors 51 to 53, the solenoid 46, and the like via a signal line or the like. Note that the respective motors 54 to 59 are connected to the control unit 100 and are individually driven and controlled by receiving individual control signals from the control unit 100.

[0067] The CPU 101 is a processor that executes a computer program to execute various types of data processing and predetermined control. The RAM 103 is a computer-readable volatile or non-volatile storage device. The RAM 103 temporarily stores the computer program to be executed by the CPU 101, data output or referred to by the CPU 101 when executing various types of processing, and the like. The ROM 102 is a non-volatile storage device that stores a control program such as a BIOS and an OS in advance for causing the CPU 101 to execute various types of arithmetic processing. The storage unit 104 is a flash memory that stores various types of information. The storage unit 104 stores the control program for executing various types of processing by the control unit 100, and data, a threshold value, a reference value, and the like to be used for various types of processing. Note that the storage unit 104 may be a non-volatile storage device such as an HDD and an SSD, which is connected directly to the control unit 100 or indirectly to the control unit 100 via the Internet or the like.

[0068] Incidentally, in the image forming apparatus 10, when a sheet is taken from the sheet receiving unit 11A or the sheet housing unit 27 and fed, the sheet slips on the pick-up roller or the feed roller in some cases. The slipping of the sheet causes a delay in the conveyance of the sheet. For example, when the conveyance of the sheet is delayed during the execution of a continuous printing job in which images are sequentially formed on the sheets conveyed continuously, the sheet spacing between the leading sheet (first sheet) that is fed previously and the following sheet (second sheet) that is fed next varies. In this case, the productivity of the image forming apparatus 10, i.e., the number of sheets printed per unit time, decreases. Further, in some cases, the spacing from the further next sheet becomes extremely narrow, and there is a possibility that a sheet jam occurs in the conveying path 26 or the conveying path 70. Further, if the feed timing by, for example, the resist roller

60A is made earlier in order to eliminate the variation in the sheet spacing, there is a possibility that the inclination of the sheet is not sufficiently corrected.

[0069] In this embodiment, the control unit **100** performs the speed control of the sheet being conveyed as described below, which allows, even if a delay occurs in the following sheet that is conveyed next to the leading sheet that is conveyed previously, the delay in the sheet to be made up during the conveyance of the next sheet.

[0070] The control unit **100** controls, when the CPU **101** executes the various control programs stored in the ROM **102** or the storage unit **104** in advance, the conveying speed of the sheet conveyed from the sheet receiving unit **11**A or the sheet housing unit **27** toward the image transfer position P**0**.

[0071] As shown in FIG. **4**, the control unit **100** includes various processing units such as a resist control unit **105** and a conveying speed control unit **106**. Note that in this embodiment, not all of these processing units are necessarily essential and some of them can be omitted in some cases.

[0072] When the CPU **101** executes various types of arithmetic processing according to the control program, the control unit **100** functions as various processing units such as the resist control unit **105** and the conveying speed control unit **106**. The control unit **100** or the CPU **101** is an example of the computer or processor that executes the control program. Note that some or all of the processing units included in the control unit **100** may include electronic circuits such as a motor driver. Further, the control program may be a program for causing a plurality of processors to function as the various processing units.

[0073] The resist control unit **105** controls the rotational driving of the resist roller **60**A such that the sheet is conveyed by the sheet conveying unit **23** while the resist roller **60**A of the resist roller unit **60** is stopped to deflect the sheet whose leading edge has reached the nip portion between the resist roller **60**A and the driven roller **60**B and then the sheet is fed out at predetermined feed timing in accordance with the transfer start timing at which the transfer of the toner image to the sheet starts at the image transfer position P**0**.

[0074] The conveying speed control unit **106** controls the rotational driving of the upstream-side conveying roller **91**A and the downstream-side conveying roller **92**A for the sheet conveyed from the sheet receiving unit **11**A to control the conveying speed of the sheet conveyed through the conveying path **70**. Note that the conveying speed control unit **106** is an example of the speed control unit according to the present disclosure.

[0075] FIG. **5** and FIG. **6** are each a graph showing the relationship between the position of each sheet and the arrival time point in the case where sheets are continuously conveyed from the sheet receiving unit **11**A. In FIG. **5** and FIG. **6**, the horizontal axis indicates the position in the conveying direction and the vertical axis indicates the time. Further, in FIG. **5** and FIG. **6**, a line L**11** shown by a solid line indicates the relationship between the position of the leading edge of the leading sheet (first sheet) and the arrival time point, and a line L**12** indicted by a broken line indicates the relationship between the position of the rear edge of the leading sheet and the arrival time point. Further, a line L**21** shown by a solid line indicates the relationship between the position of the leading edge of the following sheet (second sheet) and the arrival time point.

[0076] FIG. **5** is a graph in the case where both the leading sheet and the following sheet are normally conveyed at the initial speed V**1** without slipping. Note that in FIG. **5**, a line L**22** shown by a two-dot chain line indicates the position of the leading edge and the arrival time point in the case where, for example, the following sheet is fed with a delay of a delay time X (ms) (=Δt) due to slipping or the like.

[0077] As shown in FIG. **5**, in the case where the sheet is conveyed normally, the feeding of the leading sheet starts at a time point TO, the timing at which the leading edge of the leading sheet is detected at a position P**1** by the sheet detection sensor **51** is a time point T**1**, and the timing at which the rear edge of the leading sheet is detected by the sheet detection sensor **51** is a time point T**3** later than the time point T**2**. Further, the feeding of the following sheet with predetermined set spacing Ds**1** from the leading sheet starts at the time point T**2** earlier than the time point T**3**, and the timing at which the leading edge of the following sheet is detected at the position P**1** by the sheet detection sensor **51** is a time point T**4** later than the time point T**3**.

[0078] For example, a case where the following sheet is fed at the time point T**21** later than the time point T**2** by the delay time X (ms) is considered. Hereinafter, the following sheet that is fed with a delay of the delay time X will be referred to as a delayed sheet in some cases.

[0079] In this case, when the conveyance continues at the initial speed V**1**, the leading edge of the delayed sheet follows the line L**22**. In the section up to the point where the leading edge of the delayed sheet reaches the position P**1**, whether or not the delayed sheet is delayed is unknown. However, when the leading edge of the delayed sheet reaches the position P**1** and is detected by the sheet detection sensor **51**, a time difference Δt**1** (=T**41**−T**3**) between the time point T**3** when the rear edge of the leading sheet was detected and a time point T**41** when the leading edge of the delayed sheet was detected is obtained. When the time difference Δt**1** is longer than a time difference Δt**0** (=T**4**−T**3**) in the case where there is no delay, it is possible to determine that a delay has occurred and obtain the delay time (=Δt**1**−Δt**0**). Further, actual sheet spacing Ds**2** can be obtained from the time difference Δt**1** and the initial speed V**1**, and a delay amount ΔD (=Ds**2**−Ds**1**) indicating the degree of delay can be obtained.

[0080] In this case, the delay can be made up by increasing the conveying speed (circumferential speed) of the sheet by the upstream-side conveying roller **91**A after the leading edge of the delayed sheet is detected (after the time point T**4**), and then the conveying speed of the sheet by the upstream-side conveying roller **91**A can be returned to the initial speed V**1**.

[0081] However, in this speed control, in the case where the delay amount AD is relatively large, it is necessary to rapidly accelerate the upstream-side conveying roller **91**A and then rapidly decelerate it in order to make up for the delay before the leading edge of the delayed sheet reaches the second conveying roller pair **92**, and overshooting and undershooting of the conveying speed occur, which makes the conveying speed of the sheet unstable. Further, there is also a possibility that the sheet is damaged due to the rapid change in speed. Meanwhile, if the priority is given to the stability of sheet conveyance and the prevention of sheet damage, the delay cannot be made up before the leading edge of the delayed sheet reaches the second conveying roller pair **92**.

[0082] On the other hand, in this embodiment, as shown by a line L**23** in FIG. **6**, the conveying speed control unit **106** measures, when the rear edge of the leading sheet that is conveyed previously (first sheet according to the present disclosure) is detected at the position P**1** by the sheet detection sensor **51**, a first elapsed time Ta (first elapsed amount according to the present disclosure) from the time point T**3** (first time point according to the present disclosure). The conveying speed control unit **106** then determines, in the case where the count value of the first elapsed time Ta has reached a reference time required for the sheet with the set spacing Ds**1** to be conveyed at the initial speed V**1** (reference amount according to the present disclosure), whether or not the leading edge of the following sheet (second sheet according to the present disclosure) has been detected at the time point (hereinafter, referred to as a reference time point). In the case where the leading edge of the following sheet has not been detected at the reference time point, the conveying speed control unit **106** then changes the conveying speed of the following sheet by the upstream-side conveying roller **91**A from the initial speed V**1** to a high speed V**2** faster than the initial speed V**1**. At this time, the conveying speed control unit **106** maintains the conveying speed of the sheet by the downstream-side conveying roller **92**A at the initial speed V**1**.

[0083] Note that in FIG. **6**, the line L**23** indicating the position of the leading edge of the following sheet and the arrival time point in the case where the conveying speed of the following sheet delayed during feeding has been changed by the conveying speed control unit **106** is shown.

[0084] In this regard, in the case where the conveyance of the following sheet is delayed, the conveying speed of the sheet by the upstream-side conveying roller **91**A is changed to the high speed V**2** before the leading edge of the following sheet is detected.

[0085] Since the reference time is the time Δt**0** (=T**4**−T**3**) required for the sheet with the set spacing Ds**1** to be conveyed at the initial speed V**1** and the arithmetic processing time of the speed change by the conveying speed control unit **106** is extremely short and negligible, the reference time point can be regarded as substantially the time point T**4** shown in FIG. **6**, i.e., the time point T**4** at which the rear edge of the following sheet is detected by the sheet detection sensor **51** in the case where no delay has occurred.

[0086] Further, the conveying speed control unit **106** determines, on the basis of a second elapsed time Tb (second elapsed amount according to the present disclosure) from the reference time point to a time point T**40** (second time point according to the present disclosure) at which the leading edge of the following sheet is actually detected, return timing (time point T**42**) at which the conveying speed is returned from the high speed V**2** to the initial speed V**1**.

[0087] Specifically, when an estimated continuation time (estimated continuation amount according to the present disclosure) for continuing to convey the sheet at the high speed V**2** after the reference time point is represented by Z, the conveying speed control unit **106** determines, as the return timing, the time point obtained by adding the estimated continuation time Z to the reference time point (time point T**4**).

[0088] Since the distance (=V**2**·Z) when the sheet is conveyed at the high speed V**2** for the estimated continuation time Z is equal to the distance (=V**1** (X+Z)) when the sheet is conveyed at the initial speed V**1** for the delay time X (ms) and the estimated continuation time Z, the following relational expression (1) holds true.

$$V2 \cdot Z = V1(X+Z) \tag{1}$$

[0089] Further, since the distance when the sheet is conveyed at the high speed V**2** for the second elapsed time Tb is equal to the distance when the sheet is conveyed at the initial speed V**1** for the delay time X (ms), the following relational expression (2) holds true.

$$V2 \cdot Tb = V1 \cdot X \tag{2}$$

$$X = V2 \cdot Tb / V1$$

[0090] By substituting the above formula (2) into the above formula (1) to rearrange the estimated continuation time Z, the following calculation formula can be derived.

$$Z = V2 \cdot Tb / (V2 - V1) \tag{3}$$

[0091] In the situation where the leading edge of the delayed sheet has not actually been detected, the second elapsed time Tb is unknown and thus, the calculation formula (3) is a linear function with the second elapsed time Tb as a variable. In this embodiment, the conveying speed control unit **106** calculates, at the time point T**40** when the leading edge of the delayed sheet was actually detected by the sheet detection sensor **51**, the estimated continuation time Z using the second elapsed time Tb and the calculation formula (3) and determines, as the return timing, the time point T**42** obtained by adding the estimated continuation time Z to the reference time point (time point T**4**).

[0092] When it is determined that the elapsed time from the reference time point (time point T**4**) has reached the return timing (time point T**42**), the conveying speed control unit **106** then returns the conveying speed of the sheet by the upstream-side conveying roller **91**A from the high speed V**2** to the initial speed V**1**.

[0093] As a result, as shown in FIG. **6**, a position P**2** of the leading edge of the delayed sheet that has been delayed matches, at the time point T**42**, the position of the leading edge of the following sheet in the case where there was no delay, and the delay of the following sheet is made up.

[0094] Since the conveying speed of the sheet by the upstream-side conveying roller **91**A is adjusted in this way, it is possible to make up for the delay of the sheet before the delayed sheet reaches the second conveying roller pair **92**, without rapidly accelerating or rapidly decelerating the upstream-side conveying roller **91**A.

[0095] Note that in the case where the leading edge of the following sheet is detected by the sheet detection sensor **51** before the first elapsed time Ta reaches the reference time, this means that the following sheet is conveyed earlier than normal due to the positional deviation during feeding, double feeding, or the like. In this case, the conveying speed control unit **106** changes the conveying speed of the following sheet by the upstream-side conveying roller **91**A from the initial speed V**1** to a low speed V**3** lower than the initial speed V1 and returns the conveying speed to the initial speed V1 after the sheet is conveyed at the low speed V3 by the amount of the time or distance corresponding to the earlier conveyance.

[Conveying Speed Control Processing]

[0096] An example of the procedure of conveying speed control processing executed by the control unit **100** will be described below with reference to FIG. **7** and FIG. **8**, and a conveying speed control method according to the present disclosure will be described. In FIG. **7** and FIG. **8**, S**11**, S**12**, . . . indicate the numbers (Step numbers) of the processing procedure.

[0097] Note that one or a plurality of Steps included in the conveying speed control processing described below may be omitted as appropriate. Further, the order of execution of Steps in the conveying speed control processing may differ as long as the same operation and effect are achieved. Further, although a case where one processor corresponding to the control unit **100** executes the processing of each Step in the conveying speed control processing will be described below as an example, a plurality of processors may execute the respective Steps in the conveying speed control processing in a distributed manner.

[0098] The conveying speed control processing described below is executed in the case where a continuous printing job in which the sheet in the sheet receiving unit **11**A is continuously conveyed and an image is printed on the sheet has been input.

[0099] As shown in FIG. **7**, in Step S**11**, the control unit **100** drives the second feeding unit **80** to start feeding the first sheet (hereinafter, referred to a leading sheet) from the sheet receiving unit **11**A. Further, in Step S**12**, the control unit **100** causes the upstream-side conveying roller **91**A and the downstream-side conveying roller **92**A to be driven to rotate to convey the leading sheet at the initial speed V**1**.

[0100] In Step S**13**, the control unit **100** determines whether or not the leading edge of the first leading sheet has been detected by the sheet detection sensor **51**. In the case where the leading edge of the leading sheet has not been detected and a predetermined time has elapsed in this state, a timeout occurs and the control unit **100** determines that there is a sheet conveyance error (S**14**). In this case, the control unit **100** aborts the processing and performs error processing of outputting an error message (S**15**), and then, the series of processing ends.

[0101] Meanwhile, in the case where the leading edge of the leading sheet is detected by the sheet detection sensor **51**, the sheet spacing of the set spacing Ds**1** from the rear edge of the leading sheet is provided (S**16**), the control unit **100** starts feeding the next sheet (hereinafter, referred to as a following sheet) (S**17**). Note that in the case where the leading edge of the leading sheet has been detected, the control unit **100** performs conveyance by the predetermined amount obtained by adding the set spacing Ds**1** to the remaining length to the rear edge and starts feeding the following sheet at the timing when the leading sheet was conveyed by the predetermined amount.

[0102] In the next Step S**18**, conveying speed adjustment processing of adjusting the conveying speed of the upstream-side conveying roller **91**A is performed.

[0103] Specifically, as shown in FIG. **8**, the control unit **100** determines whether or not the first elapsed time Ta from the time point T**3** when the rear edge of the leading sheet was detected has reached the reference time before the leading edge of the following sheet is detected by the sheet detection sensor **51** (S**181**).

[0104] In the case where it is determined in Step S**181** that the first elapsed time Ta has reached the reference time before the leading edge of the following sheet is detected, this means that a delay has occurred in the conveyance of the following sheet. In this case, the control unit **100** increases the conveying speed of the following sheet by the upstream-side conveying roller **91**A from the initial speed V1 to the high speed V2 (S**182**).

[0105] After that, the control unit **100** calculates, after the leading edge of the following sheet is detected, the estimated continuation time Z using the calculation formula (3), and determines, as the return timing, the time point obtained by adding the estimated continuation time Z to the reference time point (time point T**4**) (S**183**).

[0106] In the case where it is determined that the elapsed time from the reference time point (time point T**4**) has reached the return timing (time point T**42**) (S**184**), the control unit **100** then returns the conveying speed of the sheet by the upstream-side conveying roller **91**A from the high speed V2 to the initial speed V1 (S**185**).

[0107] Meanwhile, in the case where it is not determined in Step S**181** that the first elapsed time Ta has reached the reference time before the leading edge of the following sheet is detected, the control unit **100** determines, in the next Step S**186**, whether or not the leading edge of the following sheet has been detected before the first elapsed time Ta reaches the reference time (S**186**). In the case where the leading edge of the following sheet has been detected before the first elapsed time Ta reaches the reference time, this means that the following sheet is conveyed earlier than normal due to the positional deviation during feeding, double feeding, or the like. In this case, the control unit **100** decreases the conveying speed of the following sheet by the upstream-side conveying roller **91**A from the initial speed V**1** to the low speed V**3** lower than the initial speed V**1** (S**187**). Note that the following sheet may be temporarily stopped for the time corresponding to the earlier arrival without decreasing the conveying speed.

[0108] After that, the control unit **100** calculates, on the basis of the initial speed V1 and the time difference between the time when the rear edge of the leading sheet was detected and the time when the leading edge of the following sheet was detected, the sheet spacing at the time point when the leading edge of the following sheet was detected, and calculates, on the basis of the low speed V**3**, the time required for the calculated sheet spacing to reach the set spacing Ds**1**. The control unit **100** then determines, as the return timing, the time point obtained by adding the required time to the time point at which the leading edge of the following sheet was detected (S**188**).

[0109] In the case where it is determined that the elapsed time from the time point when the leading edge of the following sheet was detected has reached the return timing (S**189**), the control unit **100** then returns the conveying speed of the sheet by the upstream-side conveying roller **91**A from the low speed V**3** to the initial speed V**1** (S**185**).

[0110] Note that in the case where it is determined in Step S**186** that the leading edge of the following sheet has not been detected before the first elapsed time Ta reaches the reference time, i.e., the timing at which the first elapsed time Ta reaches the reference time and the timing at which the leading edge of the following sheet is detected are substantially the same, the conveyance of the following sheet is not delayed or earlier and the following sheet is being conveyed with appropriate sheet spacing. In this case, the conveying speed by the upstream-side conveying roller **91**A is not adjusted and is maintained at the initial speed V**1**.

[0111] After that, as shown in FIG. **7**, the control unit **100** determines whether or not the following sheet is the final sheet (S**19**). In the case where the following sheet is not the final sheet, the processing returns to Step S**17** and the processing of Step S**17** and subsequent Steps is repeated. Further, in the case where the following sheet is the final sheet, the series of processing ends.

[0112] As described above, in this embodiment, since the above-mentioned conveying speed control processing is performed, even if the following sheet is delayed, it is possible to make up for the delay of the following sheet before the following sheet reaches the second conveying roller pair **92** on the downstream side and adjust the sheet spacing to the set spacing Ds**1** without rapidly accelerating or rapidly decelerating the upstream-side conveying roller **91**A. Further, in the case where the following sheet is conveyed earlier, the following sheet is temporarily decelerated. In this case, too, it is possible to adjust the sheet spacing to the set spacing Ds**1**.

[0113] Note that although an example of feeding a sheet from the sheet receiving unit **11**A has been illustrated in the above-mentioned embodiment, the same conveying speed control processing can be applied to the conveying roller **23**A of the sheet conveying unit **23** in the case where sheets are continuously fed from the sheet housing unit **27**.

[0114] Further, although an example of using an elapsed time as the first elapsed amount and the second elapsed amount according to the present disclosure has been described in the above-mentioned embodiment, for example, in the case where the conveying motors **58** and **59** are each a motor that is driven by being applied with a pulse signal as in a stepping motor and has the rotation angle (rotation speed) proportional to the number of pulse signals (number of Steps), the number of Steps for the conveying motors **58** and **59** may be applied instead of the elapsed time.

[0115] Further, the image forming apparatus **10** has been illustrated as an example of the image forming apparatus according to the present disclosure and the sheet conveying device according to the present disclosure in the above-mentioned embodiment. However, the present disclosure can be considered as a sheet conveying device including the upstream-side conveying roller **91**A, the downstream-side conveying roller **92**A, and the control unit **100**.

[0116] It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

1. A sheet conveying device that sequentially conveys a plurality of sheets, comprising:

a first conveying roller that is driven to rotate by receiving driving force from a first drive unit to convey a sheet;

a second conveying roller that is provided downstream of the first conveying roller in a sheet conveying direction and is driven to rotate by receiving driving force from a second drive unit to convey the sheet;

a sheet sensor that is provided between the first conveying roller and the second conveying roller and detects a leading edge and a rear edge of the sheet; and

a speed control unit that controls a conveying speed of the sheet by each of the first conveying roller and the second conveying roller,

the speed control unit being configured to

change, where a leading edge of a following second sheet has not been detected at a reference time point at which a first elapsed amount from a first time point at which a rear edge of a first sheet that is fed previously is detected reaches a reference amount corresponding to predetermined sheet spacing, the conveying speed of the sheet by the first conveying roller from an initial speed to a high speed faster than the initial speed, and maintain the conveying speed of the sheet by the second conveying roller at the initial speed,

determine, on a basis of a second elapsed amount from the reference time point to a second time point at which the leading edge of the second sheet is detected, return timing at which the conveying speed of the sheet by the first conveying roller is returned from the high speed to the initial speed, and

return the conveying speed of the sheet by the first conveying roller from the high speed to the initial speed at the determined return timing.

2. The sheet conveying device according to claim **1**, wherein

the speed control unit is further configured to calculate an estimated continuation amount Z from the reference time point using the following calculation formula and determine, as the return timing, a time point obtained by adding the estimated continuation amount Z to the reference time point,

$$Z = V2 \cdot Tb/(V2 - V1)$$

(wherein, V**1** represents the initial speed, V**2** represents the high speed, and Tb represents the second elapsed amount).

3. An image forming apparatus, comprising:

the sheet conveying device according to claim **1**,

a toner image being transferred to a sheet conveyed by the sheet conveying device to an image transfer position.

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